



OFFICE HOW TO

MX60 Mobile Mapping



Trimble Business Center 2026.10 · Trimble MX60

DOCUMENT

MX60 Office How To

SYSTEM

Trimble Business Center 2026.10 · Trimble MX60

SET

Technical Manual · Standard Operating Procedure · Field How To · Office How To

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OFFICE HOW TO

MX60 Mobile Mapping

How to process MX60 data in Trimble Business Center, in the order you do it. Every section answers four questions: what to do, what to look at, what to expect, and what should make you stop.

TRIMBLE BUSINESS CENTER 2026.10 · TRIMBLE MX60

Read “Stop if” in every section. Three mistakes cost more than all the others and all three are silent: registration does not change the point cloud until **Update Scans** runs; a good RMS does not prove the work succeeded; and **Cleanup cannot be undone**.

CAUTION

Data loss, an irreversible operation, a lost result, or safety

IMPORTANT

Gets the job wrong if ignored

IN PLAIN LANGUAGE

What the section means, why it matters, what to remember

TRIMBLE METHOD

Trimble documents this —
NOT NECESSARILY A REQUIREMENT

SET BY THE PROJECT

No single company-wide answer. The project decides, and records it

NO PUBLISHED FIGURE

No published number exists. Judgement, and the evidence on the job

1 How to Use This Guide

This is the operating path for a survey employee who understands survey office work but may have little MX60 specific processing experience. The goal is not to teach survey theory. The goal is to let the processor follow the MX60 workflow, recognize when the result is not normal, troubleshoot the common causes, and stop before a bad state is carried into the next step.

1.1 The four questions

Every section answers the same four questions, in the same order. If you read nothing else in a section, read **Stop if**.

Do	The clicks, in order
Look at	What to inspect once you have done it — not "check it worked", but where to point your eyes
Expect	What normal looks like, so you can tell when it isn't

Stop if

What means you do not proceed. **These are not suggestions.** Stopping costs an hour; not stopping has cost a remobilisation

Each of the four is numbered — §16.1 *Do*, §16.5 *Stop if* — because "Stop if" appears in every section and is not an address on its own. Cite the number when you report a problem with this guide: **Office How To §16.5**, not "the stop-if in registration".

1.2 The order

The guide follows the processing sequence. §2 to §12 get you from a disk to a point cloud you can look at. §13 to §21 register it. §22 to §26 check it and fix what is fixable. §27 to §34 identify, export, record and archive it. §35 is what to check when something is wrong.

You will not use every section on every job. Calibration (§13) is periodic. LiDAR QC (§24), degraded-GNSS remedies (§25) and PFIX (§26) apply only when the trajectory needs them.

1.3 Three things to know before you start

These are the mistakes that cost the most, and all three are silent. The explanation for each is in the Technical Manual at the reference given.

1	Registration does not change the point cloud. Until you run Update Scans , the cloud is the unregistered one — and it exports perfectly happily	§19
2	A good RMS does not prove the work succeeded. A bad one proves it failed. Trimble says this in identical words in two places	§23
3	Cleanup cannot be undone	§28

1.4 Who is telling you — the authority key

This guide cannot require anything. Everything in it is somebody else's instruction, and the marker says whose. That matters because the two are not the same: a Parametrix practice is a recommendation, while **a Trimble instruction and an equipment limit bind regardless.**

MARKER	WHO SAYS SO	FORCE TODAY
[TRIMBLE]	Trimble states it as a requirement, in the cited topic or manual page	Binding. Does not wait on a Parametrix decision
[TRIMBLE METHOD]	Trimble documents the method, but does not state it as a requirement	Strong advice. The method is Trimble's; the obligation is not
[EQUIPMENT]	A hardware limit or an irreversible software operation	Binding. It is a fact about the tool
[SOP §n]	A Parametrix requirement, at that clause	As strong as that clause — check its state
[RECOMMENDED]	Recommended by this project	Not company policy. Do it unless told otherwise, and record it if you do not
[TESTING · Tn]	Depends on a result nobody has yet	An interim posture, not a rule
[DECISION · D-n]	Parametrix has not decided	Raise it. Do not improvise a standing rule

[TRIMBLE] and [TRIMBLE METHOD] are not the same thing. Where Trimble states a rule — *"A visual check is needed"* — the marker is **[TRIMBLE]**. Where it documents a way of working without making it mandatory, the marker is **[TRIMBLE METHOD]**. This guide does not promote the second into the first.

This is why §1.1 says "Stop if" is not a suggestion. A *Stop if* whose authority is **[TRIMBLE]** or **[EQUIPMENT]** is not negotiable today. A *Stop if* marked **[RECOMMENDED]** is this project's recommendation — you may proceed past it, and if you do, **record that you did and why**. Either way the instruction stays direct, because a hedged instruction gets ignored.

1.5 When a section says a decision is open

A **SET BY THE PROJECT** marker means the SOP identifies a requirement whose answer is not set. You still have to do something today. Where this guide suggests what, it is marked as a suggestion and it is not a Parametrix standard.

1.6 Record as you go

Several sections end with **Record**. Those entries are the SOP's required records (SOP Appendix B), and five of them have **no software artefact behind them** — if you do not write them down at the time, nothing else will.

Templates are in **Appendix F**.

1.7 First supervised processing job

For a processor with minimal MX60 experience, use the guide in workflow order:

1. Intake and identify the mission with §§2–4.
2. Set the project and coordinate system with §§5–7.
3. Process the trajectory and generate scans with §§8–12.
4. Follow the registration path in §§13–21. Do not skip **Update Scans**.
5. Perform QC and any justified remedies with §§22–26.
6. Identify the final state, export, document and archive with §§27–34.
7. If anything does not match the expected state in a section, go to §35 before experimenting.

Do not troubleshoot by stacking adjustments or changing several settings together. Establish the symptom, check the last known good stage, make one controlled change, and record what happened.

2 Data Intake

Before anything else happens. Nothing in this section involves TBC.

2.1 Do

1. **Copy, do not move.** The source disk stays the source until a verified copy exists in two places
2. **Verify the copy** — file count and total size at minimum; a checksum comparison if your tooling allows
3. **Open the** `.mxdb` **in a scratch TBC project** — this is the definitive test that the copy worked. **A file count can look right, the copy can seem fine, and the** `.mxdb` **still be truncated**
4. **Confirm** `POS_1/raw/` **is present and non-empty**
5. **Confirm base station data** is in `Base/`, if a local base was occupied
6. **Take the raw-data backup now**, before any processing
7. Only then is the source disk available for reuse

2.2 Look at

The mission folder, which should look like this:

```
TMX<serial>-<mission id>/
├─ Base/.YYo.YYn.YYg base RINEX, if collected
├─ Camera_1 ... Camera_4/ imagery
├─ Laser_1, Laser_2/ raw scanner data
├─ POS_1/raw/ posl_*.000,.001 ... ← the irreplaceable one
├─ Extcal.json the calibration file
├─ <mission>.mxdb the file you import
└─ <mission>_*.log
```

2.3 Expect

Tens to hundreds of gigabytes. `POS_1/raw/` holds a numbered series of `posl_*` files, not one file. `Extcal.json` is small and is there.

2.4 Stop if

- `POS_1/raw/` **is missing or empty.** There is no post-processed trajectory without it and the mission is NAV-only. Raise it now, while re-collection is still a small decision
- The copy does not verify
- The mission folder has arrived without the field record

SET BY THE PROJECT

The transfer, verification and backup procedure — the six steps above are this guide's recommendation (*SOP §11.2*).

CAUTION

Do not clear a data disk until a verified copy exists in at least two locations and the `.mxdb` has been confirmed to open.

A cleared disk is not recoverable, and a mobile mapping mission is not re-drivable at reasonable cost.

Take the backup before processing, not after. Processing writes into the project and, with **Backup SBET Next to MXDB** enabled (§8), into the raw data folder beside the `.mxdb`. A backup taken afterwards is a backup of a partly processed state (*SOP §11.3*).

2.5 Record

Transfer performed, verified how, by whom, when. Where the backup is.

3 Checking Mission Information

Do this immediately after import (§7), before any processing. Seven checks, none taking a minute.

3.1 Do

Open the mission properties and the **Mobile Mapping** node in Project Explorer and check:

#	CHECK	AGAINST
1	Project coordinate system	The control network and the client requirement
2	Covered distance	The field record
3	Run count	The field record
4	Active trajectory — is it SBET, not NAV?	Intent
5	Capture Devices — the sensors listed	The configuration: Camera 3 Back Down , Camera 4 360° , and the two lasers
6	Base station data present in the project	Whether you will process the trajectory in house
7	Calibration state recorded	§4

3.2 Look at

The **Capture Devices** node, and the **Sbet** node under each run.

3.3 Expect

The tree looks like this — **note that scans will hang off the trajectory, not off the run**:

```

Mobile Mapping
├─ <mission id>
│   └─ Capture Devices Camera 3 Back Down, Camera 4 360°, Laser Left, Laser Rig
│       └─ Run 0, Run 1, Run 2 ...
│           └─ Sbet the imported trajectory

```

3.4 Stop if

- **Run count is lower than the field record.** Data was lost in transfer. Do not process it — go back to the copy
- **A sensor is missing from Capture Devices.** It was disabled or it failed in the field. That is a field-record and re-collection question, not a processing one
- **Covered distance is materially short** of what the crew logged
- **The trajectory is NAV, not SBET,** and you have no recorded reason

SET BY THE PROJECT

What is verified at import, and by whom? The seven checks above are this guide's recommendation (*SOP §12.2*).

Intake does not fix anything. If a check fails, record it and raise it (*SOP §12.4*). A processor who quietly corrects a coordinate system mismatch at intake has removed the evidence that field and office disagreed.

3.5 Record

Each check, with its result. Anything that failed, and what was done.

4 Calibration-State Intake

Do this at intake, not later. A later Cleanup (§29) removes the objects that would produce the report.

4.1 Do

1. **Copy** `Extcal.json` **out of the raw mission folder** into the project record, named with the system serial number and the mission date
2. Run the **Mission Report** — (*TBC 23991_1*)
3. Archive the report into the project record

4.2 Look at

The report's **Capture devices** table. It carries per-sensor **boresight installation** and **boresight refinement**, and a **date of calibration**.

4.3 Expect

A date of calibration you can point at, for each sensor.

4.4 Stop if

- **The calibration date cannot be established.** Which calibration a mission was processed against is a question somebody will ask later, and the answer has to exist now
- The calibration date is older than whatever interval Parametrix has set

SET BY THE PROJECT

The recalibration interval and its triggers — including whether daily removal of the Sensor Unit counts as disturbing the calibration (*SOP §15.2*).

That dated record is the only one found anywhere in the workflow (*Technical Manual §30*). There is no other place the software tells you when the system was last calibrated.

4.5 Record

`Extcal.json` and the Mission Report, both in the project record. Which calibration this mission was processed against.

5 Project Setup

5.1 Do

1. Create the VCE project
2. **Set the coordinate system, datum, epoch and geoid model — before importing anything**
3. Record what you set, and who set it

5.2 Look at

The project's coordinate system properties, against the control network and the written client requirement (SOP §6.1).

5.3 Expect

A coordinate system that matches the control you are going to register against. TBC 2026.10 ships **Coordinate System Database v115**; selecting a predefined geoid model now enters the vertical datum name automatically (*TBC RN 2026.10*).

5.4 Stop if

- The project's accuracy requirement is not stated in writing
- The CRS, datum or epoch has not been decided (*SOP §6.2*)
- Grid or ground has not been agreed with the client (*SOP §6.3*)

CAUTION

Set the project coordinate system before importing the mission.

Changing it afterwards is possible in TBC generally, but by then every derived product — scans, registrations, exports — was computed in the previous frame. **Treat it as irreversible in practice.**

Practically: if you discover the CRS is wrong after processing, the cheapest honest route is a new project and a re-import, not a change in place.

5.5 Record

CRS, datum, epoch, geoid, and who set them.

6 Coordinate Systems

This is not a lesson in datums. It is the four places mobile mapping treats them differently (*Technical Manual §12*).

6.1 Do

#		
1	Set the CRS before import	\$5
2	Set the epoch deliberately if the datum is time-dependent	TBC 2026.10 allows a non-default epoch and warns it "is intended for experienced users" (<i>TBC RN 2026.10</i>)
3	After trajectory processing, read the SBET filename	\$10
4	At export, know whether you are producing grid or ground	\$30

6.2 Look at

The SBET filename once the trajectory exists:

FILENAME	WHAT POSPAC DID
<code>sbet_[mission].out</code>	Computed directly in the project's datum and epoch
<code>sbet_[mission]_[frame].out</code>	Computed first in ITRF00, then transformed into the project datum and epoch

(*TBC 25943*)

6.3 Expect

Either form. Neither is an error.

6.4 Stop if

Nothing here stops you on its own. But **record which form you got**, because the second means an additional transformation happened that nobody chose.

The filename is an indicator, not a verdict. It tells you a transformation occurred. It does not tell you the parameters were right or that the result is accurate — and the plain form is equally not proof of correctness (*Technical Manual §12.3*).

NO PUBLISHED FIGURE

Which of Parametrix's normal coordinate systems POSPac recognises directly, and which trigger the ITRF00 path. Answerable once, then known.

6.5 Record

The SBET filename, in full, in the delivery record (§28).

7 Importing the Mission

7.1 Do

1. Confirm the project CRS is already set (§5)
2. Import the `<mission>.mxd`
3. If you will process the trajectory in house, **import the base station observation file** from `Base/` — the `.YYo`

7.2 Look at

Project Explorer, under **Mobile Mapping**. Then go straight to §3 and do the seven checks.

7.3 Expect

A mission node, a **Capture Devices** node, and one node per run with an **Sbet** trajectory beneath it.

7.4 Stop if

- The `.mxd` will not open. Go back to the copy (§2) — this is the definitive test that the transfer worked
- The run count or covered distance disagrees with the field record (§3)

CAUTION

Import the observation file only. The `Base/` folder also holds `.YYn` and `.YYg` ephemeris files. Trimble states: **"Do not import the ephemeris files into TBC."** (TBC 25943)

Scans are children of a trajectory, not of a run. That is why a run can end up with two sets of scans that look identical in plan, and it is the structural fact the whole of §27 rests on (*Technical Manual §5.3*).

7.5 Record

Import date, and that the seven intake checks were done.

8 Trajectory Processing

Computes the SBET inside TBC, without going out to POSPac. **Requires a POSPac MMS 8.6+ licence** — see §9 if you do not have one.

8.1 Do

1. Import the base station `.YYo` first (§7)
2. Right-click the **Mission** node ► **Process Raw Trajectory Data**
3. TBC loads the raw POS data automatically and fills the settings from what the vehicle logged. **Check them — do not accept them unread.** Trimble's own instruction: *"Double check the settings values and if needed modify them"* (TBC 25943)
4. Set **Computation Mode** — **IN-Fusion+ Single Base** or **IN-Fusion+ PP-RTX**
5. **Verify the antenna model** — see below
6. Enable **Backup SBET Next to MXDB**
7. Consider **Generate QC Report**
8. **Compute**

8.2 Look at — the settings, with Trimble's defaults

SETTING	VALUES	DEFAULT
Computation Mode	Single Base · PP-RTX	<i>none stated</i>
Initialization Mode	VNAV Gyro-compassing · GAMS · GAMS and Gyro-compassing	Gyro-compassing
Multipath	Low Medium · High	Medium
Antenna Manufacturer / Type	Read from the RINEX	—
GAMS	lever arm and standard deviation	<i>dimmed if GAMS was disabled at acquisition</i>
DMI	lever arm, SD, scale factor, scale factor SD	<i>dimmed if DMI was disabled at acquisition</i>
LiDAR QC (Refine with scans)	on/off	Off — see §24

(TBC 25943)

8.3 The one field to check every time

CAUTION

The antenna model must read `Trimble 112735` for an MX60.

`Tallysman/33-3970` is the MX9 and MX50 antenna (*TBC 25943*).

It is read automatically from the RINEX, **which means it can be read wrongly**. An incorrect antenna model puts a **systematic antenna-height and reference error** into the trajectory, and nothing downstream will attribute the symptom to its real cause.

8.4 Expect

A trajectory that computes without error, and an SBET file in the project folder.

8.5 Stop if

- The antenna model is not `Trimble 112735`
- A GAMS or DMI pane is dimmed and you expected the sensor to be fitted.
Dimmed means the sensor was disabled during acquisition and logged nothing (§35). That is a field problem — unless the sensor is not fitted at all, which is still an open question on this system

(; SOP §6.4)

- The DMI scale factor came from a manual rather than a measured wheel, and the standard deviation is still at the 5 % default. Trimble's escape hatch is to **set it to 100 % if the value is not known at all** (*TBC 25943*)

NO PUBLISHED FIGURE

Whether **Multipath = Medium** is right for open-sky Parametrix corridors, and whether the DMI 5 % default is an accuracy claim anybody verified (*SOP §13.1*).

8.6 Record

The settings used, the computation mode, and the frame-and-epoch log that **Backup SBET Next to MXDB** writes. That log is the only artefact anywhere in the workflow that records the frame and epoch a trajectory was computed in, and it lives beside the raw data rather than inside a project that may later be cleaned up (*SOP §13.2*).

IN PLAIN LANGUAGE

What this section means. Recomputing the vehicle's path from the raw GNSS and inertial data, forwards and backwards, to get a better answer than the system could produce in real time.

Why it matters. The trajectory is the job. Every point in the cloud is placed relative to it, so the quality of the trajectory sets a ceiling on the quality of everything downstream. No amount of later processing raises that ceiling.

Remember this. The post-processed result (SBET) is what survey work uses. If you end up working from the real-time trajectory instead, that is a fact about the deliverable and has to be recorded, not quietly accepted.

If this is skipped or done on the wrong input. Everything after it is built on a path that is not where the vehicle actually went — and the cloud will look completely normal.

9 POSPac Requirements

9.1 What the licence gates

ROUTE	NEEDS POSPAC?
Process Raw Trajectory Data inside TBC	Yes — POSPac MMS 8.6 or later, licensed (<i>TBC 25943</i>)
Generate POSPac Position Fixes (PFIX) — §26	Yes
Processing in POSPac externally and importing the SBET	Yes, obviously
Using the SBET the vehicle or a bureau produced	No
LiDAR QC — §24	Not stated by Trimble. See below

9.2 Do — if you have the licence

Confirm the version is 8.6 or later and that TBC can see it. **Support ► License Manager.**

9.3 Do — if you do not

1. Obtain a post-processed SBET from whoever holds a licence, and import it
2. Or, if no post-processed trajectory is available at all, **stop and raise it** — see below
3. Record which route was used, on every job

9.4 Stop if

- **The only trajectory available is the NAV solution** and the deliverable is survey-grade. NAV is the real-time solution computed in the vehicle. It is a fallback, not an option

(*Technical Manual §17.2*)

SET BY THE PROJECT**Does Parametrix hold a POSPac MMS 8.6+ licence, and where is it installed?**

Without it, two of the three degraded-GNSS remedies disappear: reprocessing with different settings, and PFIX (*SOP §13.1*).

NOT DOCUMENTED BY TRIMBLE

Whether **LiDAR QC** has its own POSPac dependency. Trimble does not state one, but it is an Applanix technology and the topic directs configuration questions to the Applanix Support Team (*TBC 28972*).

10 Reading the Trajectory

Do this before you generate a single scan. It is the highest-value, lowest-effort check in the whole workflow, it takes seconds, and it tells you where everything else is going to be difficult.

10.1 Do

1. **Mobile Mapping ► Trajectory Settings ► Rendering Settings ► RMS values**
2. Set the ranges and colours you want. **The settings persist between projects** (*TBC 27248*)
3. Look at the trajectory in **Plan View**
4. For detail: **Mobile Mapping ► Reports ► Trajectory Plots**

10.2 Look at

Three things, in this order:

Where the solution degraded	That is where control is worth most (§14), and where registration will struggle
How long each degraded stretch is	A short gap bracketed by good data is bridged well. A long one is not
Whether the degradation is at the ends of the mission	The ends are where the smoother had data on one side only. If the last stretch is a different colour from the rest, that is the closing sequence talking (<i>Technical Manual §14.4</i>)

10.3 Expect

Mostly one colour, with the degraded stretches where the mission plan predicted them — under the overpass, through the tree cover — and short.

10.4 Stop if

- A degraded stretch is long and you have **no control bracketing it and no overlapping pass**. You have no remedy for it, and that is a conversation to have now rather than after registration (§25)
- The whole trajectory is degraded. Something is wrong upstream — check §8 and §35

Segments that have been registered render as "Undefined RMS" (*TBC 27248*). A registered trajectory no longer matches its `smrmsg` file, so the colouring drops out. That is a side effect, and §27 turns it into a useful one: it makes the extent of a registration visible in plan, including where a **Local** adjustment stopped.

10.5 Record

A screen capture of the RMS-coloured trajectory. It is a required QC record (SOP §16.8) and it is one click.

11 Generate Scans

This is where the trajectory meets the measurements. Before it, the scanner data is ranges and angles from a moving sensor; after it, every return has a coordinate.

11.1 Do

1. Select a **run** in Project Explorer — **one run first**, not the mission
2. **Generate Scans** from the context menu
3. Set the **Filters**
4. Set **Colorization**
5. Run it, inspect the result (§12), and only then repeat at **mission** level

11.2 Look at — the filters

FILTER	REMOVES
Range Min / Max	Returns outside a distance window
Isolated Points	Points with too few neighbours
Fog	Returns caused by fog
Sun	Returns caused by direct sunlight on the sensor
Reflective Panels	"the noise before and after a target"
Dust	Airborne dust returns (<i>Product Bulletin, January 2025</i>)

Then the **Results of Scan Generation** dialog, which records per run the filters applied, the range, and the counts.

11.3 Expect

Scans appearing **beneath the trajectory node**, not beneath the run. Generating at mission level processes all runs; at run level, one.

11.4 Stop if

- **The deliverable is sign or retroreflectivity work and Reflective Panels is on.**
Those returns may be the deliverable
- The filter set is not the one you intended. Regenerating is cheap in effort and expensive in time; getting it right on one run first is why step 1 says one run

SET BY THE PROJECT

Are scans generated coloured by default? (*SOP §13.3*)

NO PUBLISHED FIGURE

Colouriser camera preference — forward versus backward — and its effect on fringing at feature edges. Untested.

NO PUBLISHED FIGURE

Filter defaults are untested against Parametrix work. : whether **Reflective Panels** removes legitimate retro-reflective returns from signs and line marking (*SOP §13.3*).

The MX60 has no MTA stage. If you find TBC documentation about configuring a GPU driver for MTA range-ambiguity correction (*TBC 23856*), it does not apply to this system — that is the MX9 and MX90 path (*Technical Manual §5.1*).

11.5 Record

Capture the Results of Scan Generation into the project record. It is the only artefact that states which filters produced a given cloud (*SOP §13.3*).

12 Checking the Scans

A first look, on one run, before you commit to the mission.

12.1 Do

1. Open the run's scans in **3D View**
2. Set rendering to **Scan Color** — one colour per scan
3. Look along the corridor, and then at a cross-section

12.2 Look at

Coverage	Does the cloud run the full length you expected?
Both lasers	Left and right should both be present
Density at range	Thinning with distance is normal and expected
Obvious voids	Occlusion by a vehicle, or a filter that removed more than you meant
The filters' effect	Compare against an unfiltered generation if you are unsure

12.3 Expect

A clean, dense cloud. **It will look clean even when the trajectory was poor** — that is the whole problem with mobile mapping data (*Technical Manual §3.2*). This check is for coverage and filter sanity, not for accuracy.

12.4 Stop if

- A laser is missing
- Coverage is materially shorter than the run
- A filter has visibly removed something you need

This is not the QC pass. The QC pass needs control, registration and a cutting plane, and it is §22. This is a five-minute sanity check before you spend an hour generating a whole mission.

13 Calibration

Periodic, not per-job. Run it when the interval or a trigger says so (SOP §15.2), not because a dataset looks wrong.

13.1 The laser scanners

13.1 Do

1. Collect the calibration mission: **four runs — two along one road forward and backward, two along a crossing road forward and backward** (*TBC 24886*)
2. Import it, and **generate scans** from all four runs
3. **Calibrate Laser Scanners**
4. Work **both run pairs**. The dialog presents one pair at a time and the second is easy to miss
5. Read the results, then **do the visual check**

13.2 Look at

	REQUIREMENT
Crossing angle	As close to 90° as possible, tolerance ± 30°
Minimum run length	At least 20 m each side of the crossing
Ideal run length	80 m — 40 m each side
Façades	Present in each direction , in quantity
Vegetation	A few or none

(*TBC 24886 / 20716*)

Then, in the results: **Overall Overlap %**, **Overall RMS**, and per-pair RMS in **tangential, orthogonal and vertical**.

13.3 Expect

Three similar-sized components. **One much larger than the other two points at a specific part of the solution**: tangential at timing or along-track scale, orthogonal at heading, vertical at pitch or the height component (*Technical Manual §23.3*).

13.4 Stop if

CAUTION

"Good RMS values do not mean that the calibration succeeded. A visual check is needed. On the other side, bad RMS values mean that the calibration failed." (TBC 24886, 25096)

Do not accept a calibration on RMS alone. Bad numbers end it; good numbers do not finish it.

Also stop if the site has no façades in one direction, or heavy vegetation. **Façades are the measurement** — a boresight error shows as the same flat surface appearing twice when scanned from opposing directions. Vegetation is noise in exactly that comparison.

13.2 The cameras

13.5 Do

- 1. Generate scans from at least one run
- 2. Select a camera under **Capture Devices ► Manual Camera Calibration**
- 3. Pick a position on the trajectory in Plan View; the camera view displays there
- 4. Enter **Heading, Pitch or Roll** and press Enter — **the view updates immediately**

INPUT	STEP
Arrow up / down · mouse wheel	± 0.001°
Ctrl + arrow · Page Up/Down · Ctrl + wheel	± 0.01°

(TBC 24868)

13.6 Expect

A visual, iterative alignment, not a computed adjustment. You nudge until the imagery lines up with the cloud. There is no residual, and the quality is whatever care you took.

13.7 Stop if

You cannot get the imagery to sit on the cloud at more than one location. That is not a boresight you can nudge out.

SET BY THE PROJECT

Where is the calibration site, and who maintains it? (SOP §15.3)

13.3 Afterwards

13.8 Do

1. **Export the calibration JSON and archive it outside the TBC project**, named with the system serial number and the calibration date (SOP §15.5)
2. Record the calibration: date, site, who, and the result **including the visual check**

The JSON is the complete calibration state in one small file. It imports into any later project and is the only portable record of what the system's angles were on a given date. A cleanup (§29) or a lost workstation should not take it with them.

14 Importing Control

14.1 Do

1. Obtain the control file — **Shape, ASCII or CSV** (*TBC 22905*)
2. Confirm it is in the project's coordinate system (§5)
3. Import it
4. Confirm the points appear in **Plan View** and under the **Points** node

14.2 Look at

The points, in plan, against the trajectory and the scans. Do they fall where the field record says they should?

14.3 Expect

Every point you surveyed, in the right place, in the right frame.

14.4 Stop if

- The points plot in the wrong place — that is a coordinate system or units problem, and it is much cheaper to fix now than after a registration
- **The control does not bracket the extent you intend to deliver.** A **Local** registration does not adjust beyond the outermost control point (§15, §16)
- **Nobody has designated which points are control and which are independent checks.** That designation is made **before** registration, by someone other than the person registering

(SOP §7.3)

A mobile mapping control point has to be findable in the point cloud at the density and incidence angle the vehicle produced — which is a different requirement from occupiable with a prism. A painted stop-bar corner is excellent horizontally and poor vertically; a survey nail in asphalt is below cloud resolution (*Technical Manual §22.5*).

14.5 Record

The control file used, and the control-versus-check designation with the name of whoever made it.

15 GCP and Check Point Configuration

Three independent choices per point. This is the most consequential configuration in the office workflow and it is three checkboxes.

15.1 Do

For each point, in the **Control Points** list of a registration command:

SETTING	MEANING
Use XY	The point constrains the adjustment horizontally
Use Z	The point constrains the adjustment vertically
As Check	The point is paired and measured, but excluded from the adjustment

They are per-point and per-component. A point can be **Use XY** and **As Check** in Z — used horizontally, held out vertically.

15.2 Look at

The designation you were given (§14), and set it. **Do not decide it here.**

15.3 Expect

A mixture. A painted road-surface mark is usually Use XY and not Use Z. A point held out entirely is a validation point.

15.4 Stop if

- **Every point is set As Check.** TBC will not compute — the adjustment has nothing to fit
- **No point is set As Check.** Then nothing measures the result, and the residuals you are about to read measure only how well the adjustment fitted observations it was given (§20)
- **You are about to change an As Check setting during processing.** Stop. That designation was fixed before registration began, and changing it is a Project Surveyor decision that gets recorded and requires the registration to be recomputed from the imported trajectory using

Edit (§19) — not layered on top (*SOP §7.3*)

SET BY THE PROJECT

Is the control/check designation fixed before registration and unchangeable during it? Recommended at **SOP §7.3**.

The failure this prevents. A conscientious processor registers a mission, finds one check point with a larger residual than expected, and adds it to the adjustment to bring it in. Every step is well intentioned. The result is an adjustment with **no independent check at all**, and a set of residuals that now measure nothing (*Technical Manual §22.4*).

15.5 Record

Point ID, Use XY, Use Z, As Check — for every point, **outside TBC**. The software does not appear to report it (§20, §28).

16 Register a Run

One run, against surveyed control.

16.1 Do

1. In **Project Explorer**, select a run
2. Generate its scans if not already done (§11) — **the command is dimmed without at least one generated scan**
3. Import the GCP file (§14)
4. **Mobile Mapping ► Processing ► Register a Run**
5. Accept the default **Registration Name** (*RunName* Trajectory) or enter one. **This name goes to the computed trajectory — it is what you will be identifying months later (§27)**
6. Choose a **Registration Type** — see below
7. Select a GCP under the **Points** node ► **Add Selection to Control Points**
8. Set **Use XY, Use Z, As Check** for that point (§15)
9. Optionally **Activate Limit Box** — hides everything outside a box, "to remove potential parasitic points over the target"
10. Optionally **Activate Target-Bundle Adjustment** — see below
11. Select the point in the **Control Points** list. It centres in Plan View and **Point Cloud Smart Picking** opens
12. Pick the target, read the residuals, **Validate**
13. Repeat for further points
14. **Compute**. The adjusted trajectory draws **blue**; the original stays **green**
15. **Apply**

(TBC 22905)

The 30 m rule · [TRIMBLE]

"The distance in a pair of points cannot exceed the allowed maximum distance of **30 meters**" (TBC 22905). If a pick is further than that from its GCP, it is not a valid pair — pick a feature nearer the control point, or the pairing is refused.

16.2 Look at — Registration Type

TYPE	WHAT IT DOES
Global	A shift of the whole trajectory, without rotation
Local	Interpolates between control points

TYPE	WHAT IT DOES
Global, and then Local	Global first, then Local

CAUTION

Local does not extrapolate. Trimble states it is *"not for systematic error along the run or for adjusting outside the ground control points set"* (TBC 22905).

Beyond the first and last control point the trajectory is not adjusted, **and nothing indicates where the adjustment stopped.** Control must bracket the extent you intend to deliver.

NO PUBLISHED FIGURE

Which type, when. No selection rule is published, and **Global-then-Local** appears in every Trimble screenshot with no guidance attached (SOP §13.4).

16.3 Look at — Target-Bundle Adjustment

The name reads backwards. Checked = **250 m** intervals, which is **coarser**.

Unchecked = **70 m** (TBC 22905).

NO PUBLISHED FIGURE — TEST BOTH STATES AGAINST INDEPENDENT CHECKS.

16.4 Expect, after Apply

(TBC 22905)

- An **adjusted trajectory node** beneath the run, beside `Sbet`
- A new SBET on disk: `sbet_<date>_reg_####.out`, incrementing with each registration
- Picked targets renamed *RunName TrajectoryGCPName*, updated ones carrying a trailing `*`
- Trajectory properties reading `Origin: Registration result`, `Input trajectory: Imported trajectory`, and `Registration type:`

Those four properties and the numbered SBET file are your provenance record

(§27).

16.5 Stop if

- The command is dimmed — you have no generated scan on the run
- You are registering a run that has already been registered. **That stacks adjustments.** Use

Edit (§19)

- The residuals on your check points are not what you expected. Read §20 before doing anything about it

16.6 Record

Registration name, type, the trajectory node produced, and the SBET filename **with** its `_reg_####` **number**.

IN PLAIN LANGUAGE

What this section means. Adjusting the trajectory so the point cloud lands on surveyed control, by pairing each ground control point with a target you pick in the cloud.

Why it matters. This is where the data acquires its absolute position. Which points you let the adjustment use, and which you hold back as checks, decides whether you can demonstrate the result afterwards or only assert it.

Remember this. Read the residual *before* you validate a pick, not after — the software shows it live. Choose the registration type deliberately: **Local** does not adjust anything outside the outermost control point and does not tell you where it stopped. And a good RMS does not mean it worked; Trimble says so explicitly, and asks for a visual check.

If this is skipped or rushed. A registration that reports small residuals and has pulled the cloud to the wrong place — because a target was picked on the wrong surface, or because every control point was used and nothing was left to check against.

17 Register a Mission

A set of runs at once, **with every GCP reusable**. For ordinary corridor work this is probably the command you want.

17.1 Do

The sequence is that of §16, run from the mission rather than a run. What differs:

	REGISTER A RUN	REGISTER A MISSION
Scope	One run	A set of runs
GCP reuse	Once	Every GCP, as many times as it appears
Control list	One row per GCP	One row per GCP <code>instance</code>
Instances	—	One per 250 m scan section whose bounding box contains the GCP
Side	—	An instance binds to the left, right, or both sides of the scan section
Default name	<i>RunName</i> Trajectory	Reg
Target naming	<code>RunName</code> <code>TrajectoryGCPName</code>	<code>RegistrationName_GCPName_RunName</code>
Output node	<code>Reg. Trajectory</code> under the run	<code>RegTrajectory</code> under each involved run

(TBC 26473)

17.2 Look at

The **Control Points** list. One painted mark visible in four passes produces **four instances**, and each is picked separately. Unused instances are removed from the list when you compute.

17.3 Expect

Four runs adjusted against the same observation, and therefore mutually consistent — which is the point of the command. A corridor driven in both directions twice is the ordinary case.

17.4 Stop if

- You expected instances and got none. The GCP is outside every 250 m scan section's bounding box
- You are about to re-register a mission that has already been registered (§19)

SET BY THE PROJECT

Is Register a Mission the corridor default, with Register a Run reserved for single-run cases and for repairing one run in an otherwise accepted mission? And what happens to a mission registration when one run is later re-collected? (*SOP §13.4*)

17.5 Record

As §16, plus which runs were included.

18 Register Run to Run

Cloud-to-cloud, against a fixed Reference Run. It uses no surveyed control.

That is the whole point and the whole limitation: it makes two runs agree with each other. It cannot make them agree with the ground.

18.1 Do

1. Import the missions
2. **Mobile Mapping ► Processing ► Register Run to Run**
3. Enter a **Registration Name** — "This name will be given to all computed trajectories"
4. Select a pair: one **Run to Adjust**, one **Reference Run**. They may come from the same mission or from different missions. Runs list as `<mission ID last two digits> - Run <n>`
5. **Swap Runs** inverts the two
6. **+** adds the pair to the batch
7. In Plan View the **Run to Adjust draws green**, the **Reference Run draws red**
8. Repeat for further pairs. **-** removes; the arrows reorder — **TBC registers the pairs in the order given**
9. Set the **Update Scans** option — see below
10. Check **Open Cutting Plane View**
11. **Compute**

(TBC 25096)

18.2 Look at — the Results tab

RMS in **tangential, orthogonal and vertical**, every **20 m**, plus `No overlap` where the two runs do not overlap.

`No overlap` rows are information, not noise. They tell you where the comparison had nothing to compare.

18.3 Update Scans is inside this command

This is the only place the two steps merge. The **Update Scans** checkbox regenerates the Run to Adjust's scans inline (TBC 25096). Everywhere else, Update Scans is a separate command (§21).

18.4 Expect

- A trajectory beneath the **Run to Adjust**, named `GivenName: Runname_X To Runname_X+1`
- RMS statistics in the Results tab
- A cutting plane per pair, named `MissionID Last Two Digits - Run to Adjust`

18.5 Stop if

- **You have not registered to surveyed control first.** Run-to-run improves *relative* agreement. It cannot establish absolute position, and if the Reference Run is itself displaced, run-to-run will faithfully propagate that displacement into the run you adjusted
- **You have not re-checked against independent check points afterwards.** The Run to Adjust has moved; its residuals against control have changed

The order that matters — [SOP §14.6]:

1. Register the mission to surveyed control (§17)
2. Assess against independent check points (§20) and visually (§22)
3. **Only then** use run-to-run, choosing as Reference Run the pass with the better GNSS conditions and the better residuals
4. **Re-check against the independent check points**

Step 4 is the one that gets skipped.

CAUTION

The same principle as §13 applies here, in Trimble's identical wording: **good RMS does not mean the registration succeeded; bad RMS means it failed; a visual check is needed (TBC 25096).**

NO PUBLISHED FIGURE — how much run overlap is enough.

18.6 Record

Registration name, the pairs and their order, the RMS statistics, and the check-point residuals **after** the adjustment.

19 Editing a Registration

19.1 The problem

CAUTION

Getting this wrong stacks adjustments on adjustments. A processor who registers, dislikes the residuals, and registers again has applied a second correction on top of the first. The residuals will look better. The trajectory has been bent twice against the same control.

To improve a registration, use Edit. To start over, edit and Reset.

Running a registration command a second time on a run that has already been registered adjusts the **adjusted** trajectory, not the imported one — and no number in the result shows it.

19.2 Do

1. Select the **registered trajectory node**
2. **Edit** from the context menu (*TBC 25362, 26578*)
3. The command reopens with the control points, their Use XY / Use Z / As Check states and the picked targets reloaded
4. Change what needs changing
5. **Compute**, then **Apply**

19.3 Look at

That the reloaded state is the one you expect — particularly the **As Check** settings. Edit is also how you confirm what a previous registration actually used, since no report of it has been found (§20).

19.4 Expect

The registration recomputed **from the imported trajectory**, not from the adjusted one.

19.5 Stop if

- **A reload prompt appears and you are about to answer "No".** See below
- You cannot find the Edit command. Do not fall back to running the registration again

CAUTION

If Registration Auto-Saving is on, picked targets are written to `Targets.csv`. Reopening the command prompts to reload them, and *"if you choose 'No', they will be emptied from the Targets.csv file and you will not be able to retrieve them"* (TBC 22905).

`Targets.csv` holds the registration's observations. **Answering "No" discards the field book.**

NO PUBLISHED FIGURE — whether Registration Auto-Saving is on by default. One glance at the dialog answers it.

19.6 Record

That the registration was edited rather than repeated, and what changed.

20 Residual Review

20.1 Do

1. Read the residuals in the **Targets** pane and in the **Validate Picking** window
2. Separate them: **residuals on points used in the adjustment** and **residuals on points held As Check**
3. Write both into the control-and-check table (Appendix F)
4. Read the three-axis breakdown where you have one

20.2 Look at — what TBC gives you, at four levels

LEVEL	INDICATOR	WHERE
Per pick, live	Easting, Northing, Elevation residual to the GCP, signed ; RMS of the fitted plane	Validate Picking, Targets pane (<i>TBC 22905</i>)
Per run pair, every 20 m	RMS in tangential / orthogonal / vertical; No overlap	Results tab, Register Run to Run (<i>TBC 25096</i>)
Whole calibration	Overall Overlap %, Overall RMS, per-pair three-axis	Calibrate Laser Scanners (<i>TBC 24886</i>)
Trajectory-wide	Position, orientation, velocity RMS after smoothing	Plan View colouring (§10)

20.3 Look at — the three axes, when you have them

DOMINANT COMPONENT	POINTS AT
Tangential — along travel	Timing, or along-track scale. Check the DMI scale factor and time synchronisation
Orthogonal — across travel	Heading. The hardest component, and the one a single direction of travel cannot resolve
Vertical	Pitch, or the height component. Check the antenna model (§8) and the geoid

Three similar-sized components mean random disagreement, which is what good data looks like. **One much larger than the other two is the solution telling you which part of itself is struggling** (*Technical Manual §23.3*).

20.4 Expect

Residuals on the **used** points to be small. That is not evidence of anything — an adjustment with few observations fits them exactly, and one whose observations share a systematic error fits them beautifully and carries the error straight through.

20.5 Stop if

CAUTION

"Good RMS values do not mean that the calibration succeeded. A visual check is needed. On the other side, bad RMS values mean that the calibration failed." (TBC 24886, 25096)

A number can prove failure. A number cannot prove success.

Stop if:

- **There are no As Check residuals.** Nothing is measuring the result
- A check-point residual is much larger than the used-point residuals, and you are about to promote it into the adjustment. **Do not.** Read §15 again
- One three-axis component dominates. Diagnose it before accepting

What TBC does not report. No captured topic produces a statement of **which points were used as control and which as checks** in a registration already applied. The state is visible while the command is open and reloads on **Edit** (TBC 25362, 26578) — but no report of it has been found.

NO PUBLISHED FIGURE .,. TBC 2025.21 says signed residuals are "included in the report" without naming it, and the only mobile mapping report topic does not mention residuals. Ten minutes with the software answers it.

20.6 Record

The control-and-check table: point ID, Use XY, Use Z, As Check, and the residual on each. **Six columns, written once.** It is the single most important record in the workflow and the software does not produce it (§28, SOP §20).

21 Update Scans

The step that is easiest to skip and most expensive to skip.

21.1 Do

1. Select the run or mission
2. **Update Scans**
3. Choose the **registered** trajectory
4. Confirm the result in Project Explorer — see **Look at**

21.2 Look at

The scan stations. Updated ones carry a `_reg_####` suffix:

```
Run_14_Laser Right_reg_0001 (S3)
```

And their position in the tree: they hang beneath the **registered trajectory**, not beneath `Sbet`.

21.3 Expect

A **second** set of scans, beneath the adjusted trajectory. The original set is still there, beneath `Sbet`, and both look identical in plan. That is not a duplicate to tidy away — they are the same raw data computed against two different trajectories (*Technical Manual §5.3*).

21.4 Stop if

- The stations do not carry `_reg_####`
- The scans still sit beneath `Sbet`

CAUTION

Registration does not modify the point cloud until Update Scans is performed.

An operator can complete a registration, obtain good residuals, accept the result, and then export point cloud data that still reflects the pre-registration trajectory. **The export succeeds. The file is valid. The data is unregistered.**

Update Scans works both ways. It switches between the imported and the adjusted trajectory, so it is also how you revert.

One exception: **Register Run to Run** has an **Update Scans** checkbox inside the command (§18). That is the only place the two steps merge.

21.5 Record

That Update Scans was run, against which trajectory. The pre-export check in §31 confirms it again before anything leaves.

IN PLAIN LANGUAGE

What this section means. Registration computes a new trajectory. **Update Scans is what moves the point cloud onto it.**

Why it matters. Until this runs, the project contains an adjusted trajectory and an unadjusted cloud sitting next to each other. Everything looks finished. Measurements taken from that cloud are taken from the unregistered data.

Remember this. After Update Scans, the station names gain a suffix — that suffix is how you tell by looking that the cloud in front of you is the registered one. Check it before you export.

If this is skipped. You deliver the unregistered cloud while believing you delivered the registered one. This is the single most common way this workflow goes wrong, and nothing in the software warns you.

22 Visual QC

This is the layer that catches what numbers cannot. It has no software artefact, so if you do not record it, there is no evidence it happened.

22.1 Do

1. Point Clouds ► View ► Cutting Plane View
2. Set rendering to Scan Color — see below
3. Drag the plane along the corridor, full length. It is tedious, and the tedium is the check — error arrives in stretches, so a plane dropped in three places finds nothing
4. Work the checklist under **Look at**
5. Record what you covered, and by whom

22.2 The setting that makes or breaks this

IMPORTANT — [TRIMBLE] — THE VISUAL CHECK ITSELF IS TRIMBLE'S INSTRUCTION (§20).

[SOP §16.5] — what the pass covers is this project's checklist.

Set rendering to Scan Color — one colour per scan. Without it, two offset surfaces read as one thick surface and the exact defect this check exists to find is invisible (*Technical Manual §25*).

It is part of the method, not a display preference.

22.3 Look at

CHECK	LOOKING FOR
Overlapping passes in the cutting plane, full length	Doubled surfaces
Flat surfaces at range — a wall, a building face	Thickening with distance: attitude error or a calibration issue
The ends of the corridor	Where a Local adjustment stopped; where the smoother was weakest
The degraded stretches you found in §10	Whether the registration actually fixed them
Vertical surfaces against horizontal	Systematic tilt
Features near control versus far from control	Residual growth with distance from constraint

(SOP §16.5)

22.4 Expect

Overlapping passes landing on each other. Flat surfaces that stay flat as range increases.

22.5 Stop if

- Two passes are visibly offset from each other anywhere
- A wall thickens with range
- The cloud is good near control and degrades between — that is the shape of an adjustment that fitted its constraints and nothing else

SET BY THE PROJECT

What does a visual point-cloud QC pass cover? The checklist above is this guide's recommendation (*SOP §16.5*).

NO PUBLISHED FIGURE — the working cutting-plane thickness for these checks.

SET BY THE PROJECT — how much of a corridor is inspected, and how that is decided (*SOP §16.5*).

22.6 Record

That the visual check was performed, by whom, and over what extent. No software artefact exists. This is the record.

IN PLAIN LANGUAGE

What this section means. Actually looking at the data — in profile, across two passes, at features you know the shape of — rather than only reading numbers about it.

Why it matters. Trimble states it directly: good RMS values do not mean the work succeeded, and a visual check is needed. Numbers can prove failure. They cannot prove success.

Remember this. Two passes that agree with each other can both be wrong together, if the trajectory was drifting through that stretch. Agreement is not accuracy. Check against something independent as well as against the data's own consistency.

If this is skipped. A dataset passes on statistics and fails on the ground, and the failure is found by whoever builds on it rather than by us.

23

Imagery QC

23.1 Do

- 1. Step through the imagery along the corridor
- 2. Work the checklist
- 3. Check alignment against the point cloud at a feature edge

23.2 Look at

CHECK	LOOKING FOR
Coverage	Gaps where a camera stopped, or a run was not colorized
Exposure	Blown highlights; blocked shadows under canopy and in underpasses. Both unrecoverable
Motion blur	Speed too high for the light available
Obstruction	Aerials, a following vehicle, a smear on the dome
Focus and contamination	Rain, dust, insects on the optical surface
Corrupted images	Silent — see below
Alignment with the cloud	Colour in the wrong place at feature edges: a camera boresight issue (§13.2)

(SOP §16.6)

23.3 Expect

Resolution by configuration (TBC 22501, 23888):

	CORE	PRO	PREMIUM — OURS
Panoramic	8192 × 4096	12288 × 6144	12288 × 6144
Side / planar	4096 × 3008	4096 × 3008	4096 × 3008

This system is the Premium. Panoramas are **12288 × 6144**. A panorama that comes out at 8192 × 4096 did not come from this system (SOP §6.4).

23.4 Stop if

- **A corrupted side camera image exports as black.** It is silent: nothing warns you, and the export succeeds. If you find one, assume there are others
- Coverage has a gap you cannot account for
- The imagery does not sit on the cloud

SET BY THE PROJECT

Whether the **file-size scan** for finding silently corrupted imagery is used. It is a screening method proposed by this project and **not validated** (*Technical Manual §26*).

NO PUBLISHED FIGURE — whether exported imagery reflects a registration at all, and which imagery streams the MX60 actually has and TBC exposes.

23.5 Record

That the imagery check was performed and by whom. **No software artefact exists.**

SET BY THE PROJECT — IMAGERY PRIVACY. WHETHER UNBLURRED ORIGINALS ARE retained, and for how long. Decided before collection, not on request (*SOP §19.6*).

24

LiDAR QC

Uses the **scan data itself** as an aiding sensor to improve the trajectory — SLAM-like, run inside trajectory processing. The only degraded-GNSS remedy that needs **neither POSPac nor additional ground control**.

24.1 Before you start — can this machine run it?

	MINIMUM	RECOMMENDED
CPU	Intel Core i7 or i9	Intel XEON
RAM	128 GB	256 GB
TEMP storage	1 TB SSD on the PCI bus	2 TB SSD M.2
Virtual memory	+1 TB SSD always available	+2 TB M.2, or 4 TB combined
Paging file	—	initial = installed RAM, maximum = 6 × installed RAM
Also required	MATLAB Runtime R2024b (24.2) , installed after TBC	

(TBC 28972)

24.2 Do

1. Check **LiDAR QC (Refine with scans)** in **Process Raw Trajectory Data** (§8). A LiDAR QC tab appears
2. Select the runs from the Project Tree **with overlap** — parallel runs, or crossing runs
3. **Add**
4. Set the parameters
5. **Compute**

24.3 Look at — the settings

SETTING	VALUES	DEFAULT
Range	Minimum and maximum LiDAR measurement distance used	3 m to 100 m
Noise	5, 10, 50, 80, 100, 200 mm	5 mm for MX50/MX60
Lasers	Left · Right · All	All

24.4 Expect

A long computation. It solves the constant IMU boresight angles and corrects the post-processed trajectory, position and orientation, from voxels matched in overlapping scan regions (*TBC 28972*).

24.5 Stop if

- **The runs do not overlap.** It has nothing to match
- The machine does not meet the requirement above. It is not a setting you can push through

NO PUBLISHED FIGURE

Two defaults are untested, and one of them contradicts Trimble's own guidance printed beside it: **Lasers = All**, where the note says using both "can increase computation time without significantly improving the accuracy, as it compares the left versus right laser of isolated runs." Also the **3–100 m** range: the MX60's useful range and the range over which scan geometry usefully aids a trajectory are different questions (*Technical Manual §11.3*).

24.6 The acquisition geometry it wants

ELEMENT	REQUIREMENT
Area	A structured scene — detached houses, objects within sensor range; open sky
Strips	Two perpendicular strips , each consisting of two runs, one in each direction
Strip length	250–300 m

(*TBC 28972*)

That is materially the same geometry the laser scanner calibration wants (§13).

One site can serve both, which matters because establishing one is real work (*SOP §15.3*).

24.7 Record

That LiDAR QC was run, on which runs, with which settings.

25 Degraded GNSS

You are here because §10 showed a degraded stretch. **Four remedies, and they are not interchangeable.**

25.1 Look at — what each costs and what it needs

REMEDY	ACTS ON	NEEDS	SECTION
Reprocess with better base data or settings	The GNSS side of the solution	Raw data intact; better corrections available	§8
LiDAR QC	Adds scan data as a third aiding sensor	Overlapping runs , and a very large workstation	§24
PFIX	Injects surveyed control into a second POSPac pass	POSPac licence , surveyed control, picked targets	§26
Registration to control	Bends the finished trajectory to fit control	Surveyed control	§16, §17

The first three improve the *solution*. Registration improves the *fit*. That distinction matters: a registered trajectory has been adjusted to agree with the control it was given, so its agreement with that control is no longer evidence of anything (*Technical Manual §8.5*).

25.2 Do

1. Establish how long the degraded stretch is **in time**, not in metres
2. Check what you actually have: overlap? control bracketing it? a POSPac licence?
3. Choose. Prefer a remedy that improves the solution over one that improves the fit
4. Apply it, then **re-check against independent check points** (§20)

25.3 Expect

Trimble publishes positioning performance at **no outage** and after a **60-second outage**, and nothing beyond MX60 UG Rev B, p.56:

	CORE / PRO	PREMIUM — OURS
No outage	X,Y < 0.01 m · Z 0.01 m	X,Y < 0.01 m · Z 0.01 m
After 60 s outage	X,Y 0.12 m · Z 0.1 m	X,Y 0.1 m · Z 0.07 m

Use the Premium column. The no-outage row assumes a DMI, which this system may not carry.

25.4 Stop if

- **The outage is materially longer than 60 seconds.** Beyond the published figure you are extrapolating past the manufacturer's stated envelope. Inertial drift is not linear
- **No remedy applies** — no overlap, no bracketing control, no licence. Then the honest finding is that **mobile mapping may not be the appropriate acquisition method for that segment**, and that goes up, not into the deliverable

SET BY THE PROJECT — THE DECISION RULE WHEN A CORRIDOR PRODUCES AN unacceptable trajectory: who decides, against what, and what the client is told (*SOP §22.5*).

The remedies that need something from the field cannot be arranged now. Overlap for LiDAR QC and control bracketing a hostile stretch are mission-planning decisions (*SOP §8*). If they were not made, the option does not exist today.

25.5 Record

Which remedy, why, and the check-point residuals before and after.

IN PLAIN LANGUAGE

What this section means. What you can and cannot do about a stretch where the satellites were blocked and the trajectory drifted.

Why it matters. Only some remedies actually add information. Control in the affected stretch adds real, independent information. Registering the bad pass to a good one adds consistency but no new truth. Knowing which is which keeps you from making a dataset look better without making it better.

Remember this. If there is no overlap, no control bracketing the stretch and no reprocessing option, the honest outcome is to say the stretch does not meet the requirement — not to smooth it until it looks acceptable.

If this is skipped. A degraded stretch is delivered inside an otherwise good dataset, carrying the same accuracy statement as the rest of it.

26 PFIX

Generate POSPac Position Fixes. Where registration corrects the answer, PFIX corrects the computation: the control observations go back into the navigation solver as position fixes and the filter re-solves with them available.

26.1 Prerequisites

(TBC 24460)

- **POSPac MMS installed with a valid licence** for the IN-Fusion processing methods
- An SBET processed in POSPac — or the NAV trajectory, "in case of POSPac processing not possible"
- A VCE project whose coordinate system matches the data
- The `.mxd` imported with that trajectory applied
- **Scan data generated from at least one run**
- A GCP file imported in the project coordinate system

26.2 Do — part one, in TBC

1. Right-click the **Mission** node ► **Generate Pospac Position Fixes**. *The command does not open if the mission has no generated scan*
2. Select a GCP under **Points** ► **Add Selection to Control Points**
3. Pick the target — **the same Point Cloud Smart Picking tool as registration** (§16), with the same live residuals and the same **30 m maximum pair separation**
4. **Validate**. Easting, Northing and Elevation residuals display
5. Add further pairs
6. **Compute**. Updated targets are named `Mission_Name-PFIX-GCP_Name`, and a `custom_events.txt` is written into a **PFIX** folder under the TBC project folder
7. Close the dialog

26.3 Do — part two, in POSPac

1. Start POSPac MMS, create and save a project
2. Import the POS logged files from `POS_1/raw`
3. **Copy** `custom_events.txt` **from TBC into the** `Extract` **folder of the POSPac project**
4. Open the **GNSS-Inertial Processor**
5. Optionally open **Position Fixes and Satellite Events** to inspect the fixes
6. Select the IN-Fusion processing mode
7. **All Processings.** A new SBET appears in the `Proc` folder

26.4 Do — part three, back in TBC

1. Select the mission ► properties
2. **Replace the initial trajectory file with the new SBET**
3. **Update Scans** (§21)

26.5 Expect

A better trajectory through the stretch that had no GNSS, with the correction propagated by the filter's own model of how the system behaves rather than by interpolation between control points.

26.6 Stop if

- **You stop after step 7 of part one.** The `custom_events.txt` does nothing by itself
- **You skip part three's Update Scans.** A better trajectory that never reaches the point cloud has cost a day and changed nothing in the deliverable

NOT DOCUMENTED BY TRIMBLE

When should PFIX be preferred over registration? Trimble describes both commands and never contrasts them. The framing at the top of this section is this project's reading, not Trimble's statement (*Technical Manual §27.5*).

26.7 Record

That PFIX was used, on which stretch, and the check-point residuals before and after.

27 Identifying Registration Results

Which trajectory is this cloud built on? You will be asked. This section is how you answer.

27.1 Do

Work the four layers, in this order.

#	LAYER	WHERE
1	Tree position	Scans hang beneath the trajectory that produced them
2	Station suffix	Updated stations carry <code>_reg_####</code>
3	Trajectory properties	Origin: Registration result Input trajectory: Registration type: (TBC 22905, 26473)
4	SBET filename on disk	<code>sbet_<date>_reg_####.out</code> , incrementing per registration, in the project folder

27.2 Look at

Project Explorer, expanded. A run that has been registered has **two trajectories and two sets of scans**:

```
Run 14
├─ Sbet the imported trajectory
│  └─ Run_14_Laser Right (S1) ← unregistered scans
└─ Reg. Trajectory Origin: Registration result
    └─ Run_14_Laser Right_reg_0001 (S3) ← registered scans
```

They look identical in plan. Tree position and the suffix are the difference.

27.3 A fifth, incidental indicator

Registered trajectory segments render as **"Undefined RMS"** in the RMS colouring (§10), because a registered trajectory no longer matches its `smrmsg` file (TBC 27248).

That makes the **extent** of a registration visible in plan — including where a **Local** adjustment stopped adjusting, which nothing else shows you.

27.4 Expect

All four layers agreeing.

27.5 Stop if

- **They disagree.** A cloud beneath `sbet` whose stations carry `_reg_####` is telling you something you need to resolve before exporting
- **You cannot tell which trajectory a cloud was built on.** Do not export it (§31)

Nothing in this is conclusive once the data leaves the project. That is what §28 is about.

27.6 Record

The trajectory node name and the SBET filename **with its** `_reg_####` **number**, in the delivery record.

28

Trajectory Provenance

28.1 The limitation, stated plainly

IMPORTANT

Exported mobile mapping data may retain coordinate, timing, and in some formats trajectory information, but the captured Trimble documentation does not establish that the output uniquely identifies the adjusted trajectory or registration result used to create it.

Inside the project, §27 gives you four layers of evidence. **Outside it, you have what you wrote down.**

28.2 Do

Record these, now, before Cleanup (§29) and before export (§30):

#	ARTEFACT	WHERE IT COMES FROM
1	The delivery record — mission ID, trajectory node name, SBET filename including <code>_reg_####</code> , registration type, export path and date, exported by whom	Written by you. Appendix F
2	The control-and-check table — point ID, Use XY, Use Z, As Check, residual on each	Written by you (§20). Appendix F
3	The Mission Report , run before Cleanup	(TBC 23991_1)
4	The calibration JSON in force	§13.3
5	<code>Targets.csv</code>	The project folder
6	The field record	It came with the data (§2)
7	The QC record — including the visual and imagery checks	Written by you (§22, §23)

(SOP §20.2)

28.3 Expect

Five of the seven already exist as files. Two are written by a person. The whole package is a few hundred kilobytes beside a project of tens of gigabytes.

28.4 Stop if

- You are about to run Cleanup and any of items 3 to 5 is not archived
- You are about to export and item 1 does not exist

SET BY THE PROJECT — WHAT PROVENANCE RECORD ACCOMPANIES A DELIVERABLE, WHERE

it lives, and who produces it (*SOP §20.2*).

NO PUBLISHED FIGURE — which trajectory travels with a publish or an export; what a LAS file actually carries in its header, VLRs and sidecar; and whether a delivered dataset can be matched back to its trajectory after the fact.

29 Cleanup Mobile Mapping Mission

CAUTION

Cleanup Mobile Mapping Mission is destructive and cannot be undone.

It permanently removes registrations, trajectories and scan sets from the project, keeping only the most recent. Trimble states: *"Please, have a backup copy of your project prior performing the operation, it cannot be undone."* (TBC 26466)

29.1 Do — the archive-first sequence

[EQUIPMENT] Cleanup cannot be undone. That is Trimble's own statement about the command and it binds today. **[SOP §18.3]** — the ten-step sequence, and the requirement for written authorisation, are this project's recommendation. **Who may authorise a Cleanup is and is not settled**, which is why step 8 names an authority that does not exist yet.

In this order. Steps 2 to 6 are small files; the whole set is a few megabytes.

#	STEP	WHY HERE
1	Complete and accept QC (§22, §32)	Cleanup is an end-of-preparation step. Before acceptance it removes the alternatives you might need
2	Run the Mission Report and archive it	It records capture devices, runs, trajectories, generated scans, and per-sensor calibration with date . Run it before Cleanup — afterwards it can only report what survives
3	Record the registration evidence (§20, §28)	Control/check designation and residuals. TBC does not report these
4	Archive <code>Targets.csv</code>	The picked registration observations — not recoverable
5	Archive the numbered SBET files <code>sbet_<date>_reg_####.out</code>	Pending, assume Cleanup removes them
6	Archive the calibration JSON (§13.3)	The system state the mission was processed under
7	Take the project backup Trimble asks for — into the project archive, not a local copy	A backup nobody can find is not a backup
8	Obtain the written authorisation	SOP §18.2
9	Run Cleanup	

#	STEP	WHY HERE
10	Record that it was run — by whom, on what date, what was archived first	Otherwise the absence of history is itself unexplained

(SOP §18.3)

29.2 Look at

Before running it: the project tree, so you know what you are about to lose.
Registered trajectories, their scans, and the numbered SBET files.

29.3 Expect

Only the most recent registration surviving.

29.4 Stop if

- **You do not have written authorisation** — **[RECOMMENDED]** (SOP §18.2).
Until is answered there is no appointed authoriser; raise it rather than proceeding on your own
- Any of steps 2 to 6 is not done
- The backup went to your own machine rather than into the project archive

NO PUBLISHED FIGURE

Does Cleanup delete `sbet_*_reg_####.out` **from storage, or only remove the project objects?** Until it is known, assume the worse case and archive them.

SET BY THE PROJECT — when Cleanup may be performed, by whom, and what must be archived first.

29.5 Record

Authorisation; what was archived and where; that Cleanup was run, by whom, on what date.

IN PLAIN LANGUAGE

What this section means. Cleanup permanently deletes the intermediate data a mobile mapping project accumulates, to get the project size back under control.

Why it matters. It cannot be undone, and it removes things you may need to prove later what was delivered and how it was produced.

Remember this. Copy out everything you will want afterwards *before* you run it — the trajectory file, the calibration, the registration record, the targets. Then confirm you have written authorisation to run it at all.

If this is skipped — or run too early. The deliverable survives and the evidence behind it does not. When somebody later asks which trajectory produced the delivered cloud, there is no way to answer.

30 Export

Six documented MX60 paths. No preference between them is expressed here; format choice is a project and client matter Parametrix has not decided (*SOP §19.4*).

Do §31 first. Every path below assumes the pre-export check has passed.

30.1 The paths

PATH	WHERE	CARRIES THE TRAJECTORY?
Export to LAS (Trajectory Split)	Mobile Mapping tab	No, despite the name
Export to TMX	Mobile Mapping tab	Yes — one trajectory file for all devices, in a folder under the Mission folder
Export to TopoDot	Mobile Mapping tab	The <code>.lst</code> carries image position and orientation
Export to Solv3D	Mobile Mapping tab	Sidecar <code>reference.csv</code>
Generic Point Cloud Export	Point Cloud tab	No
Publish to TRCPS	Home ► Data Exchange ► Publish to TRCPS	Yes — point cloud and trajectories are exported by default

30.2 Do — Export to LAS (Trajectory Split)

1. Run **Extract Classified Point Cloud first** — *Point Clouds ► Regions*. Without it there is nothing to export
2. **Home ► Data Exchange ► Export ► Mobile Mapping tab**
3. Set **Splitting distance** — "a value in meter multiple of 250"
4. Choose lasers: merged into one file, or left and right independently. **At least one must be selected**
5. Set **Format** — LAS 1.2 or 1.4 — and **Export unit**

(TBC 27279)

NO PUBLISHED FIGURE

Sample points performs *random* sampling to a fixed point count. On a survey deliverable that is destructive thinning with no documented spatial rule — no minimum spacing, no preservation of edges or breaklines. Its default state is not stated. **Check it before exporting.**

30.3 Do — Export to TMX

- 1. **Mobile Mapping tab ► Export to TMX**
- 2. Decide the **Export timestamps** setting — **read §31 first**
- 3. Export

Produces panoramic images, side camera images, laser point clouds **and the trajectory**, plus a `reference.csv` (TBC 22501).

30.4 Do — Export to TopoDot

- 1. **Generate scans first.** "Otherwise, nothing will be exported"
- 2. **Close all run views.** "Otherwise, a warning message will pop up"
- 3. Export

Produces LAS 1.4, one couple per run (TBC 23339).

30.5 Do — Export to Solv3D

- 1. **Mobile Mapping tab ► Export to Solv3D**
- 2. Trimble recommends **disabling timestamps** on this path

Produces a mission-named folder with `Lasers` and `Panorama` sub-folders, LAS 1.4 (TBC 23888).

On this path the recommended setting is also the safe one: timestamps off means the **generated** scans are exported rather than reprocessed ones (§31).

30.6 Do — Generic Point Cloud Export

- 1. **Home ► Data Exchange ► Export ► Point Cloud tab**
- 2. **This tab selects by region or a drawn rectangle — not by run** (TBC 11769)
- 3. Set **Scaling**: **Grid** or **Ground**, or **ECEF**

OPTION	BEHAVIOUR
Grid	Current projected CRS with the combined scale factor. Writes a <code>.txt</code> sidecar naming the coordinate system and scale factor. Trimble warns re-importing it "may cause some inconsistencies due to a double-scaling effect"
Ground	Ground coordinates scaled from the 0,0 origin. "The scale factor is not exposed during export"
ECEF	LAS or LAZ in ground-based scaling, "that includes the project's global coordinate system information"

IMPORTANT

A ground-scaled export does not record the scale factor it used. The recipient cannot recover it from the file. Agree grid or ground in writing, and make sure the delivery can say which (*SOP §6.3*).

This is the path most likely to be used for an ordinary LAS deliverable, and it is the one with the least documented provenance and no run awareness.

NO PUBLISHED FIGURE — what happens when a Point Cloud tab selection is drawn across scans belonging to two different trajectories. Not documented.

30.7 Do — Publish to TRCPS

1. **Home ► Data Exchange ► Publish to TRCPS ► Mobile Mapping tab**
2. Requires a **Trimble ID**; uploads via the **Trimble Desktop Utility**, installed with TBC
3. **Point cloud and trajectories are exported by default** — the publishing options govern imagery only (*TBC 29527*)

NOT DOCUMENTED BY TRIMBLE — · NO PUBLISHED FIGURE

Which trajectory is published when several exist under a run is not documented.

From TBC 2026.10, Trimble ID sign-in requires two-step verification — a code by email each time. Worth knowing before it stops a session.

30.8 Record

Export path, date, by whom, format, scaling — into the delivery record (§28, Appendix F).

31 The Pre-export Check

One minute. It prevents the most expensive failure in the workflow.

[EQUIPMENT] — the behaviour this guards against is Trimble's, and it is silent: an export of the unregistered cloud succeeds and produces a valid file. **[SOP §19.2]** — whether the check is *mandatory*, and whether export may proceed without it, is not yet decided.

31.1 Do

1. In **Project Explorer**, find the scan nodes you are about to export
2. Confirm they sit **beneath the intended registered trajectory**
3. Confirm their stations carry the `_reg_####` suffix
4. Screen-capture the tree
5. Decide the **Export timestamps** setting — see below
6. Only then export

(SOP §19.2)

31.2 Look at

```
Run 14
├─ Sbet ← NOT this one
│  └─ Run_14_Laser Right (S1)
└─ Reg. Trajectory ← this one
    └─ Run_14_Laser Right_reg_0001 (S3)
```

31.3 Stop if

CAUTION

Registration does not modify the point cloud until Update Scans is performed.

An operator can complete a registration, obtain good residuals, accept the result, and then export point cloud data that still reflects the pre-registration trajectory. **The export succeeds. The file is valid. The data is unregistered.**

Stop if the scans sit beneath `Sbet`, or the stations have no `_reg_####`. Go back to §21.

31.4 Export timestamps

CAUTION

With **Export timestamps** set to **Yes**, Trimble states that *"the exported scans are **reprocessed from the raw data**"* rather than being the scans processed with Generate Scans (*TBC 23339, 22501*).

Which trajectory that reprocessing uses is not stated. Until this is established, the exported data may not be the data that was registered and checked.

[TESTING ·] — an interim posture, not a rule, and it is withdrawn the day is answered in either direction. Until then: **treat an export with timestamps enabled as unverified against the checked dataset, and do not enable it on a delivered dataset without a recorded reason** (*SOP §19.3*).

SET BY THE PROJECT

Is this confirmation mandatory, and may export be performed without it? (*SOP §19.2*)

NO PUBLISHED FIGURE — the reliable export-state verification method for each path. How an export dialog resolves its selection is not documented.

31.5 Record

The screen capture, and the timestamps setting used.

32 Final QA/QC

Each layer catches something the others cannot. None is optional because another was performed.

32.1 Do

Work them in order and record each.

#	LAYER	CATCHES	WHERE	ARTEFACT
1	Field coverage verification	Missing passes, missing overlap	Field record	Field record
2	Intake checks	Transfer loss, wrong CRS, missing sensors	§3	Intake record
3	Trajectory RMS review	Where the solution was weak — before any cloud exists	§10	Screen capture
4	Residuals on control	A broken adjustment	§20	Targets pane
5	Residuals on independent checks	An adjustment that fits its own observations and is still wrong	§20	Written by you
6	Visual inspection of the cloud	Doubled surfaces, thickening at range, tilt	§22	No software artefact
7	Imagery inspection	Coverage, exposure, blur, corruption	§23	No software artefact
8	Export-state confirmation	Delivering the unregistered cloud	§31	Screen capture

(SOP §16.2)

32.2 Look at

Layer 5 especially. **Residuals on points that took no part in the adjustment are the only numerical evidence that means anything**, and they exist only if somebody designated check points before registration began (§15).

32.3 Expect

Layers 6 and 7 to produce nothing you can file unless you write it. **Two of the eight layers have no software artefact at all.** If a reviewer asks whether the visual check was performed and over what extent, the only possible answer is a record somebody wrote.

32.4 Stop if

- **There are no As Check residuals**
- Any layer was skipped because another one looked fine
- You are about to accept on RMS alone

CAUTION

"Good RMS values do not mean that the calibration succeeded. A visual check is needed. On the other side, bad RMS values mean that the calibration failed." (TBC 24886, 25096)

32.5 Acceptance is not yours

Acceptance is a decision by the person with the authority under SOP §4, recorded, against the project's stated accuracy requirement. Your job is to produce the evidence, not to conclude.

SET BY THE PROJECT

What constitutes an acceptable registration and an acceptable point cloud is not established. Trimble publishes no acceptance tolerance for the MX60 and none has been set by test. **Do not invent one, and do not quote one.**

What is blocked is formal acceptance, not the work. You can process, register and inspect a dataset with open — this section is how. What cannot happen is an acceptance resting on a Parametrix standard, because there is not one. Whoever signs is signing on their own documented judgement.

The decision put to Parametrix — whether interim acceptance against a project-specific written requirement is permitted at all — is **SOP §17.2.**

32.6 Record

The QA/QC record: every layer, who performed it, when, and over what extent. Appendix C and Appendix F.

33 Archiving

33.1 Do

1. Assemble the **record package** (§28) — seven artefacts, a few hundred kilobytes
2. Decide and apply the **retention tier** for everything else
3. Write the **archive record** — one page
4. Put it where somebody who was not involved can find it

33.2 Look at — the three tiers

Tier 1 — small, irreplaceable, keep indefinitely

Field record · control-and-check table with residuals · Mission Report run **before**
Cleanup calibration JSON · Results of Scan Generation · delivery record · QA/QC
record · the accuracy statement issued.

The whole of Tier 1 is a few hundred kilobytes. There is no storage argument against keeping it.

Tier 2 — a defined period

SBET and its processing report and the frame-and-epoch log · numbered registered
SBETs `Targets.csv` · base station data · the delivered files.

Tier 3 — per the decision

	SIZE	NOTE
Raw mission folder	Tens to hundreds of GB	The only thing that permits reprocessing. Once gone, the deliverable cannot be improved, only re-collected
TBC project	Tens to hundreds of GB	The provenance record

(SOP §21.2)

33.3 Stop if

- **You are about to delete** `POS_1/raw/` **or** `Targets.csv` **on your own judgement.** Neither is recoverable: one cannot be recomputed, the other was picked by a person and would be different if picked again (SOP §21.3)
- The archive record does not exist

SET BY THE PROJECT — WHAT IS RETAINED, WHERE, FOR HOW LONG, BY WHOM
(SOP §21.1).

SET BY THE PROJECT — folder structure, naming and storage location.

33.4 The archive record

One page per project: what was archived, where, when, by whom; the retention tier applied; whether Cleanup was run and what was archived first; and where the raw mission data is, if it is held elsewhere.

Without it, the archive is a folder somebody has to reverse-engineer.

33.5 Record

The archive record itself. Appendix F.

34 Documentation Required

One list, so nothing is discovered missing at the end. Everything here is required by the SOP; its full index is SOP Appendix B.

34.1 The package

#	ARTEFACT	FROM	EXISTS AS A FILE?
1	Intake check record	§3	No — you write it
2	<code>Extcal.json</code> and the Mission Report	§4	Yes
3	Trajectory settings and the frame-and-epoch log	§8	Yes
4	Trajectory RMS screen capture	§10	Yes, once you capture it
5	Results of Scan Generation	§11	Yes
6	Calibration record and JSON	§13	Yes
7	Control-and-check table with residuals	§20	No — you write it
8	Visual QC record	§22	No — you write it
9	Imagery QC record	§23	No — you write it
10	Update Scans confirmation	§21	Screen capture
11	Pre-export check capture and the timestamps setting	§31	Screen capture
12	Delivery record	§28	No — you write it
13	QA/QC record	§32	No — you write it
14	Cleanup authorisation and what was archived	§29	No — you write it
15	Archive record	§33	No — you write it

34.2 Look at

Eight of the fifteen do not exist unless a person writes them. That is not an oversight in the software; several of these facts have no representation in it at all.

34.3 Expect

To spend perhaps thirty minutes across a project producing all of them, most of it copying files out before Cleanup.

34.4 Stop if

You are at delivery and items 7, 8 or 12 do not exist. Those are the three a reviewer asks for first, and none can be reconstructed afterwards.

Templates for the writing-required items are in **Appendix F**.

35 Common Problems

35.1 Troubleshooting pattern

Do not troubleshoot an MX60 project by trying commands until the picture looks better.

1. **State the symptom.** Name the command, object, residual, visual condition or missing output.
2. **Find the last known good stage.** Intake, trajectory, scan generation, registration, Update Scans, QC and export are separate states. Go back to the first state that is wrong.
3. **Change one thing at a time.** Stacked registrations and repeated processing can hide the cause.
4. **Verify visually as well as numerically.** A clean number is not proof that the cloud is right.
5. **Preserve the evidence.** Do not run Cleanup or overwrite a useful intermediate state while you are still diagnosing the problem.
6. **Raise field failures as field failures.** Missing coverage, missing overlap, a dead sensor or a missing closing sequence cannot be repaired by clever office processing.

35.2 Symptom table

WHAT YOU SEE	CHECK FIRST	IF THAT IS NOT IT	STOP OR RAISE WHEN
A registration command is dimmed	Does the run have at least one generated scan ?	Return to §11 and confirm scan generation completed	The required object still is not available after the prerequisite exists
GAMS or DMI settings are dimmed in Process Raw Trajectory Data	Was the sensor enabled during acquisition?	Check the field record and Field How To §10	If it did not log, treat it as a field configuration issue, not a TBC setting to invent
Trajectory is NAV, not SBET	Is <code>POS_1/raw/</code> present and is POSPac available?	§§2, 8 and 9	The required trajectory processing path is unavailable

WHAT YOU SEE	CHECK FIRST	IF THAT IS NOT IT	STOP OR RAISE WHEN
Picked targets vanished	Was a reload prompt answered No ?	Check §19 and the state of <code>Targets.csv</code>	The target record was emptied and cannot be reconstructed reliably
Residuals got worse after a second registration	Did you register twice?	Use Edit per §19 instead of stacking another adjustment	You no longer know which adjustment state produced the current result
Exported cloud is not the registered one	Was Update Scans run?	Confirm the stations carry <code>_reg_####</code> and use §31	The export state cannot be positively tied to the reviewed registration
Update Scans was run but export still points at <code>Sbet</code> scans	Check the exact scan nodes selected for export	Re-run the §31 pre-export check and select the <code>_reg_####</code> scans beneath the intended registered trajectory	The export selection cannot be positively tied to the checked registration
Two passes are offset in the cutting plane	Is rendering set to Scan Color ?	Follow §22 and check whether the offset changes with range	The passes remain visibly inconsistent after the correct view is established
Surface looks thick even though numbers look fine	Attitude error or calibration	Inspect whether thickening grows with range; §§13 and 22	Visual QC contradicts the numerical result
Side camera images export black	Corrupted imagery	Check the rest of the image set using §23	Imagery is required and corruption is not isolated
Imagery color sits beside feature edges	Camera boresight	§13.2 and visual QC	The offset persists and affects the intended use
Nothing exports to TopoDot	Were scans generated and run views closed?	Follow §30 from its beginning	The expected objects exist but the exporter still produces nothing
Extract Classified Point Cloud produced nothing	Was it run before Export to LAS (Trajectory Split) ?	Repeat the documented sequence in §30	The documented order still does not produce the expected object

WHAT YOU SEE	CHECK FIRST	IF THAT IS NOT IT	STOP OR RAISE WHEN
A wrong antenna model is found before processing	It must read Trimble 112735	Correct the input before computing; §8	The correct MX60 antenna model cannot be established
A wrong antenna model is discovered after downstream work exists	Stop using the derived scans/registration/export as the current state	Return to §8, correct and recompute the trajectory, then rebuild the downstream products that depended on it	You cannot establish which derived products were built from the corrected trajectory
Whole trajectory is degraded	Antenna model must read Trimble 112735	Base data, coordinate information and §8	The model or trajectory inputs cannot be verified
Height bias across the whole job	Antenna model, geoid and base coordinate	§§5 and 8	The bias is systematic or its cause cannot be isolated
RMS coloring has gaps	Registered segments can show Undefined RMS	§§10 and 27	The gap is accompanied by another trajectory or registration problem
No overlap rows in run-to-run results	Confirm the runs truly overlap there	§18	Required overlap was not collected
LiDAR QC does not produce a useful result	Do the runs overlap and does the machine meet the memory requirement?	§24	The prerequisite data or hardware is not available
MTA or GPU driver documentation appears relevant	It is not the MX60 path	Return to §11	Stop using MX9/MX90 instructions on an MX60 job
TBC sign-in asks for an emailed code	TBC 2026.10 uses Trimble ID two-step verification	Follow the current sign-in path	Access cannot be established without bypassing company or Trimble controls

35.3 When the answer is "re-collect"

Missing coverage, missing overlap, a mission with no closing sequence and a sensor that logged nothing are field problems. The office guide may help identify them, but it cannot manufacture the missing observations.

Raise the non-conformance. Do not absorb it. A quiet workaround makes the current job harder to defend and teaches the next processor the wrong recovery method (*SOP §22.3*).

A Office Processing Checklist

Intake to delivery. Tick as you go; the § column is where the detail is.

A1 · Intake

☐	§
☐ Copy verified — file count and size, checksum if available	2
☐ <code>POS_1/raw/</code> present and non-empty	2
☐ Base station data present, if a local base was occupied	2
☐ Raw-data backup taken before any processing	2
☐ Field record received	2
☐ <code>Extcal.json</code> copied into the project record	4
☐ Mission Report run and archived; calibration date established	4

A2 · Project and import

☐	§
☐ Accuracy requirement stated in writing	5
☐ CRS, datum, epoch, geoid set — before import	5
☐ Grid or ground agreed in writing	5
☐ <code>.mxd</code> imported; base <code>.YYo</code> imported (not <code>.YYn</code> / <code>.YYg</code>)	7
☐ Coordinate system matches the control network	3
☐ Covered distance consistent with the field record	3
☐ Run count matches the field record	3
☐ Active trajectory is SBET, not NAV	3
☐ Capture Devices lists the expected sensors	3

A3 • Trajectory

☐		\$
☐	Antenna model reads <code>Trimble 112735</code>	8
☐	Computation mode set deliberately	8
☐	GAMS / DMI panes checked — dimmed means the sensor logged nothing	8
☐	Backup SBET Next to MXDB enabled	8
☐	Trajectory computed	8
☐	RMS colouring reviewed and captured	10
☐	SBET filename recorded — plain or <code>_frame</code>	6

A4 • Scans

☐		\$
☐	One run generated first, filters checked	11
☐	Filters appropriate to the deliverable	11
☐	Results of Scan Generation captured	11
☐	Mission generated	11
☐	First look done — coverage, both lasers, voids	12

A5 • Registration

See **Appendix B**.

A6 • QC and delivery

See **Appendix C** and **Appendix D**.

A7 · Close-out

☐		§
☐	Record package complete — seven artefacts	28
☐	Cleanup, if run, followed the archive-first sequence	29
☐	Archive record written	33

B Registration Checklist

B1 · Before you open the command

☐	\$
☐ Scans generated on the run(s) — the command is dimmed otherwise	11
☐ Control imported, in the project coordinate system	14
☐ Control plots where the field record says it should	14
☐ Control brackets the extent you intend to deliver	14
☐ Control-versus-check designation received, in writing, from the Project Surveyor	15
☐ You are not the person who made that designation	SOP §4.3

B2 · Setting it up

☐	\$
☐ Registration name chosen — this is what you will identify months later	16
☐ Registration type chosen deliberately	16
☐ Use XY / Use Z / As Check set per the designation — not per your judgement	15
☐ At least one point held As Check	15
☐ Not every point held As Check	15
☐ Target-Bundle Adjustment state decided — checked = 250 m, coarser	16

B3 · Picking

☐	\$
☐ Each pick validated, residuals read at the time	16, 20
☐ GCP-to-target separation within 30 m	26
☐ Reload prompt — never answered "No"	19

B4 · After Compute

☐	§
☐ Adjusted trajectory node present	16
☐ <code>sbt_<date>_reg_####.out</code> on disk, number recorded	16, 27
☐ Trajectory properties read: <code>Origin: Registration result</code> , input trajectory, registration type	27
☐ Residuals on As Check points recorded, by component	20
☐ Three-axis breakdown read where available	20
☐ Update Scans run	21
☐ Stations carry <code>_reg_####</code>	21, 27

B5 · If something needs changing

☐	§
☐ Edit used — never a second registration	19
☐ Any change to an As Check designation went back to the Project Surveyor and was recorded	15

B6 · Run-to-run, if used

☐	§
☐ Registered to surveyed control first	18
☐ Reference Run is the pass with better GNSS and better residuals	18
☐ Pair order set deliberately — TBC registers in the order given	18
☐ Check points re-checked afterwards	18

C

QC Checklist

Eight layers. Each catches something the others cannot. None is optional because another one looked fine.

☐	#	LAYER	EVIDENCE PRODUCED	\$
☐	1	Field coverage verification	Field record	2
☐	2	Intake checks — seven	Intake record	3
☐	3	Trajectory RMS review	Screen capture	10
☐	4	Residuals on control points used	Targets pane	20
☐	5	Residuals on independent check points	Written by you	20
☐	6	Visual inspection — Cutting Plane View, Scan Color	Written by you	22
☐	7	Imagery inspection	Written by you	23
☐	8	Export-state confirmation	Screen capture	31

C1 · Layer 6 in detail

☐	CHECK	LOOKING FOR
☐	Rendering set to Scan Color	Without it the check does not work
☐	Overlapping passes, plane dragged full length	Doubled surfaces
☐	Flat surfaces at range	Thickening with distance
☐	The ends of the corridor	Where a Local adjustment stopped
☐	The degraded stretches from layer 3	Whether registration fixed them
☐	Vertical against horizontal	Systematic tilt
☐	Near control versus far from control	Residual growth with distance

C2 · Layer 7 in detail

☐	CHECK
☐	Coverage — gaps where a camera stopped or a run was not colorized
☐	Exposure — blown highlights, blocked shadows. Both unrecoverable
☐	Motion blur
☐	Obstruction — aerials, following vehicle, smear on the dome
☐	Focus and contamination
☐	Corrupted images — silent, export as black
☐	Alignment with the point cloud at feature edges

C3 · Before you call it done

☐	
☐	As Check residuals exist and are recorded
☐	No layer was skipped
☐	Layers 6 and 7 were written down — nothing else records them
☐	Nothing is being accepted on RMS alone
☐	No acceptance tolerance has been invented or quoted — is open

D Export and Delivery Checklist

D1 · The gate

☐	\$
☐ QC complete, all eight layers	32
☐ Accepted by the person with the authority — not by you	32
☐ Scan nodes sit beneath the intended registered trajectory	31
☐ Stations carry <code>_reg_####</code>	31
☐ Tree screen-captured	31
☐ Export timestamps setting decided and recorded	31

D2 · The export

☐	\$
☐ Path chosen, and it carries what the client needs	30
☐ Grid or ground per the written agreement	30
☐ If ground: the scale factor is communicated separately — the file does not carry it	30
☐ If Trajectory Split: Extract Classified Point Cloud run first	30
☐ If Trajectory Split: Sample points checked — it randomly thins	30
☐ If TopoDot: scans generated, all run views closed	30
☐ If Point Cloud tab: the selection does not cross two trajectories	30

D3 · What goes with it

☐	\$
☐ Delivery record — mission ID, trajectory node, SBET filename with <code>_reg_####</code> , registration type, export path, date, by whom	28

☐		\$
☐	Coordinate system, datum and epoch stated in the delivery, not only in the file	30
☐	Accuracy statement, where accuracy is relied on	32
☐	Control-and-check table, where the client receives the evidence	20

D4 · After

☐	
☐	What was delivered, to whom, when, in what format and scaling — recorded
☐	The delivered files retained per the retention tier

E Archive and Cleanup Checklist

CAUTION

Cleanup Mobile Mapping Mission is destructive and cannot be undone.

It permanently removes registrations, trajectories and scan sets from the project, keeping only the most recent. Trimble states: *"Please, have a backup copy of your project prior performing the operation, it cannot be undone."* (TBC 26466)

E1 · The archive-first sequence

In this order. Do not reorder it.

☐	#	STEP	§
☐	1	QC complete and accepted	32
☐	2	Mission Report run and archived — before Cleanup, or it can only report survivors	29
☐	3	Registration evidence recorded — designation and residuals	20
☐	4	<code>Targets.csv</code> archived	29
☐	5	Numbered SBET files archived — <code>sbet_<date>_reg_####.out</code>	29
☐	6	Calibration JSON archived	13
☐	7	Project backup taken into the project archive — not a local copy	29
☐	8	Written authorisation obtained	29
☐	9	Cleanup run	29
☐	10	Recorded — by whom, on what date, what was archived first	29

E2 · Archive

☐	§
☐ Tier 1 complete — a few hundred kB, keep indefinitely	33
☐ Tier 2 stored with the retention period noted	33

☐		§
☐	Tier 3 decision applied and recorded — raw mission folder and TBC project	33
☐	<code>POS_1/raw/</code> and <code>Targets.csv</code> not deleted by an individual acting alone	33
☐	Archive record written — one page	33

E3 · Tier 1, itemised

☐	
☐	Field record
☐	Control-and-check table with residuals
☐	Mission Report, run before Cleanup
☐	Calibration JSON in force
☐	Results of Scan Generation
☐	Delivery record
☐	QA/QC record
☐	Accuracy statement issued

F

Record Templates

Five templates for the records that **have no software artefact**. Each is small. Each is unrecoverable if it is not written at the time.

F1 · Control and check table

The single most important record in the workflow, and TBC does not produce it (§20).

PROJECT		MISSION		REGISTRATION NAME	
Designated by		Date designated		Registered by	
Point ID	Use XY	Use Z	As Check	Residual E	Residual N
---	---	---	---	---	---
	☐	☐	☐		
	☐	☐	☐		
	☐	☐	☐		

Designation is fixed before registration and is not changed during it (§15, SOP §7.3). If it changes, the Project Surveyor decides, it is recorded here with a date, and the registration is recomputed using **Edit** — not layered on top.

F2 · Delivery record

One page per delivered dataset.

Project	
Mission ID	
Trajectory node name	
SBET filename, including <code>_reg_####</code>	
Registration type	Global · Local · Global then Local · Run to Run
Update Scans confirmed	☐ stations carry <code>_reg_####</code>

Export path	
Export timestamps	☐ No ☐ Yes — <i>if Yes, state the recorded reason</i>
Scaling	☐ Grid ☐ Ground ☐ ECEF · scale factor communicated: ☐
Coordinate system, datum, epoch	
Export date	
Exported by	
Delivered to	

F3 · QA/QC record

LAYER	PERFORMED BY	DATE	EXTENT COVERED	RESULT
1 Field coverage verification				
2 Intake checks				
3 Trajectory RMS review				
4 Residuals on control				
5 Residuals on independent checks				
6 Visual inspection				
7 Imagery inspection				
8 Export-state confirmation				
---	---			
Accuracy requirement assessed against				
Accepted by				
Acceptance date				

Acceptance is not the processor's. It is a decision by the person with the authority under SOP §4, against the requirement stated at project setup.

F4 · Cleanup authorisation and archive-first record

Project · Mission	
Authorised by	
Authorisation date	
Mission Report archived	☐ · location:
Registration evidence recorded	☐
<code>Targets.csv</code> archived	☐ · location:
Numbered SBET files archived	☐ · location:
Calibration JSON archived	☐ · location:
Project backup taken	☐ · location:
Cleanup run by	
Date run	

F5 · Archive record

One page per project.

Project	
Archived by · date	
What was archived	
Where it is	
Retention tier applied	☐ Tier 1 ☐ Tier 2 ☐ Tier 3
Retention period runs from	

Cleanup run?	☐ No ☐ Yes — see F4
Where the raw mission data is, if held elsewhere	
Disposal, if any — what, when, authorised by	

Without this, the archive is a folder somebody has to reverse-engineer.